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Emergent particle collection by cyanobacteria through gliding motility and filament buckling

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This manuscript presents **important** work on macrostructure formation in a freshwater filamentous cyanobacterium, focusing on its ability to aggregate and buckle. The authors employ a wide range of experiments and approaches, including time-lapse imaging and 3D theoretical modelling, to establish the physical dynamics of filaments, such as gliding motility and buckling. They demonstrate that, in addition to gliding motility, filament length and flexibility are essential for the ability of cyanobacteria to capture particles and form aggregates. Overall, the study provides **convincing** evidence and advances our understanding of the role of microbial motility in the environment. It will be of broad interest to biophysicists and environmental microbiologists.

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Abstract

Cyanobacterial macrostructures such as mats, aggregates, and stromatolites are observed in diverse habitats. These structures incorporate organic and inorganic matter and form microenvironments enabling biochemical transformations. Macrostructures are found to be associated with motile, filamentous cyanobacteria, but to what extent, and how filament motility can drive macrostructure formation is unclear. To address this question, we study macrostructure formation in a well-characterised freshwater cyanobacterial community dominated by the filamentous cyanobacterium *Fluctiforma draycotensis*. We discover an emergent particle collection behaviour that results in aggregate macrostructures composed of a solid core surrounded by an outer layer dominated by entangled cyanobacterial filaments. We show that the particle collection and subsequent aggregate formation result from the gliding motility of the cyanobacteria. Developing a novel 3D model of active filament movement, we analyse how filament morphology and mechanical properties affect movement dynamics, particle collection, and filament entanglement. We predict that particle collection requires filaments above a certain length and flexibility. We confirm the resulting prediction of length dependence of particle collection behavior with shortened filaments of *F. draycotensis* and naturally short *Pseudanabaena* sp. filaments. Together, our results show that filamentous cyanobacteria can use gliding motility to actively engineer their environments through particle collection and macrostructure formation, and that this ability is confined to a part of the filament phase space in terms of length and flexibility. These insights will allow better prediction of macrostructure formation in natural habitats and specific cyanobacterial species, and the engineering of cyanobacterial macrostructures for biotechnological applications.

Introduction

Cyanobacteria form a diversity of macroscopic structures in aquatic and terrestrial ecosystems. Aggregates and granules can form in aquatic environments, including lakes, open ocean, and bioreactors [1–6], while biofilms and mats are found on rocks or sediments in environments including hot springs and tidal flats [7]. Macrostructures often contain different bacterial species in multiple distinct layers [7–9]. The metabolic activities of resident microorganisms generate gradients of light, oxygen and redox potential, and cycling of key metabolites such as hydrogen and sulfur [4, 5, 10–15]. Therefore, cyanobacterial macrostructures form key habitats enabling microbial transformations of (in)organic matter and thought to have played a key role in the original oxygenation of the Earth’s atmosphere [7, 10, 16].

Macrostructure morphologies may be smooth or pustular, and may also generate protruding laminated structures known as stromatolites [7, 17]. Cyanobacterial mat sections transferred to the laboratory show vertical and conical growth through accumulation and burial of organics [1, 8, 17–20]. Mats can also incorporate inorganic or nutrient-rich sediment particles, aiding in sediment stabilisation [21, 22] and mineral deposition [23]. Similarly, marine cyanobacterial aggregates can attach to and incorporate air-borne dust [24]. Cyanobacteria can act as adsorbing, passive surfaces [7, 25, 26], contributing to these macrostructures through adsorption and flocculation. However, other mechanical drivers of particle incorporation and determinants of macrostructure morphology remain unknown.

Environmental macrostructures are often specifically associated with motile, filamentous species of cyanobacteria [2, 7, 24]. Genes associated with gliding motility – particularly those involved in the synthesis and secretion of motility-enabling polysaccharides – are present across nearly all filamentous cyanobacteria [27]. The extracellular polysaccharides, secreted during gliding motility and broadly known as ‘slime’, can also act as a flocculating agent and cause attachment of filaments to each other and to organic substances [25, 26]. Diverse filamentous cyanobacteria form different types of macrostructures in the laboratory [3, 4, 28], but the exact role of gliding motility in the formation dynamics of macrostructures is unclear. Therefore, the cyanobacterial factors influencing macrostructure formation and eventual morphology remain an open question [7, 8, 17] and the ‘engineering’ of cyanobacterial granules for biotechnological applications remains empirical and unpredictable [2, 5].

Here, we studied macrostructure formation in detail, using a motile filamentous cyanobacterium (*Fluctiforma draycotensis*, from the Cyanobacteriales order) that reproducibly forms macroscopic granules (‘macrostructures’) under laboratory conditions. This allowed us to track structure formation across macro- and microscopic lengthscales, and timescales from seconds to days. We discovered an emergent behaviour where filament gliding motility generates dynamic collection of solid particles, which are incorporated into granular macrostructures. Combining experiments with a novel physical model of filament propulsion and mechanics, we identified that this behaviour depends on filament length and flexibility, and showed that shortened filaments do not collect particles to form granular structures. A contrasting short motile filamentous species (from order Pseudoanabinales, whose lineage diverged from Cyanobacteriales approximately two billion years ago), also does not perform particle collection, as predicted by our model. We thus establish a clear role for gliding motility and filament physical properties in dictating macrostructure formation and morphology. These insights and future analyses of presented model systems, will enable better understanding of environmental cyanobacterial macrostructures and their engineering for biotechnologies.

Results

Macrostructure formation is driven by gliding motility and particle collection

We studied macrostructure formation first in a microbial community dominated by a motile, filamentous cyanobacterium, *F. draycotensis* (order Cyanobacteriales) [4] (see *Methods*). This species shows a form of gliding motility characteristic of many filamentous cyanobacterial species, involving surface-attached back-and-forth movement, principally along the long axis of the filament [29] (Movie S1 [\[link\]](#)). The filaments also rotate around their long axis as they move, with the rotation tightly coupled to forward translation [29]. We focused on the repeatable emergence of macrostructures in *F. draycotensis* cultures as a model system, then explored another cyanobacterial culture dominated by an alternative motile, filamentous species, *Pseudanabaena* sp. (order Pseudanabaenales). For both static and gently shaken *F. draycotensis* cultures, macrostructures formed within 1–2 days of culture initiation and persisted over the lifetime of the culture, which can be up to several months (Figure 1A [\[link\]](#) and *Methods*). Mature macrostructures formed under static conditions were highly robust to agitation by swirling or shaking, and comprised an envelope of cyanobacterial filaments surrounding a mm-scale solid yellow core that is enriched in iron compared to the media (Figures 1B [\[link\]](#) and S1). Yellow flocculate iron precipitates were observed to form in non-inoculated media, but they always remained sub-mm and readily dispersed upon minimal agitation (Figure S2 [\[link\]](#)). Macrostructures were not observed in cultures containing a mutant, non-motile *F. draycotensis*, or cultures containing only the other community members without *F. draycotensis* (Figure S2 [\[link\]](#)). These results show that macrostructure formation in this system is driven by the motility of the filamentous cyanobacteria.

To understand the formation of these granular macrostructures with embedded particles, we imaged a dilute, homogenised culture in a glass beaker of diameter ≈ 4 cm, over a period of ≈ 48 hours (Figure 1C [\[link\]](#) and *Methods*). We repeatedly observed the emergence of highly motile granular structures, which reproducibly grew to a diameter of 1 mm within 24 hours (Movie S2 [\[link\]](#)). Specifically, small yellow particles first appeared at the bottom of the beaker, often surrounded by visible bundles of cyanobacteria filaments. These particles moved with, and were actively collected by, cyanobacterial filaments (Movies S2, S3 [\[link\]](#)). Increasingly larger granular structures were formed through coalescence of two or more smaller structures, which appear yellow or green depending on the density of cyanobacteria around the structure (Figure 1D [\[link\]](#) and Movie S3 [\[link\]](#)). We quantified the size and speed of macroscopic structures over the course of a 33-hour filming period (Figure 1E [\[link\]](#) and *Methods*). The size increase over this period is mainly due to the macrostructure formation dynamics, as the biomass doubling time in *F. draycotensis* cultures is in the order of one week [4]. We find that structures of 1 mm diameter form after about 20 hours, and move at speeds of 1–4 $\mu\text{m/s}$ (Figure 1E [\[link\]](#), inset), comparable to measured gliding speeds for individual *F. draycotensis* filaments [29].

In later stages, larger structures were often visibly green due to their dense covering of cyanobacteria (Figure 1A [\[link\]](#)). Macrostructures could also become denser through collected particles and filaments ‘tightening’ (Figure 1E [\[link\]](#)). We found that particle collection by cyanobacteria was not selective for the yellow precipitate particles, and polystyrene macroscopic spheres (430 μm diameter) were similarly collected if provided, pointing to a physically rather than chemically-mediated process (Movie S4 [\[link\]](#)).

Particle collection is associated with buckling instabilities

We next investigated how particle collection happens mechanistically at the microscopic level, and which specific motility behaviours displayed by *F. draycotensis* are relevant. We took timelapse videos at the microscale, using fluorescence microscopy and microscopic polystyrene beads (48 μm diameter, see *Methods*), and observed their dynamic collection (Movie S5 [\[link\]](#)). In a representative example shown in Figure 2A [\[link\]](#), initially homogeneously distributed beads were

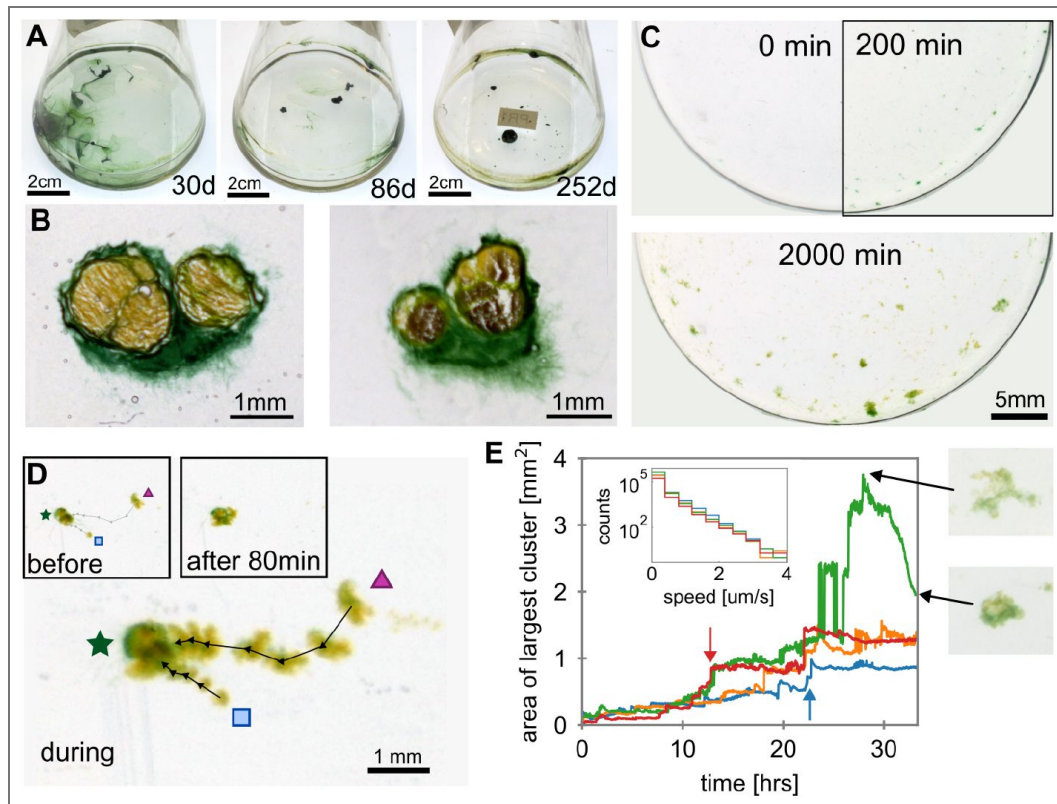


Fig. 1. Granule formation at the macroscale.

(A) Images showing cultures at 30, 86, and 252 days of growth. (B) Slices through mature granules, showing cyanobacteria in green and yellow cores containing solid precipitates. (C) Images of aggregate formation in a glass beaker, at 0, 200, and 2000 minutes (see [Movie S2](#)). (D) Time-projected, composite image showing aggregate growth through coalescence of two smaller particles (indicated by triangle and square), with a larger particle (indicated by star). Arrows show the trajectory of the small particles. Insets at top show individual images before (0min) and after (80min) the coalescence event, respectively (see [Movie S2](#) and [S3](#) for example coalescence events). (E) Maximum particle size within field of view, plotted against time. Different colours indicate data from separate experiments. Coalescence events cause visible 'steps' in the maximum particle size (red and blue arrows). An aggregate 'tightening' event is also visible in the data – see still images taken before and after 'tightening' event. Inset shows the distribution of aggregates' speed.

collected into one of two large ($\approx 100\ \mu\text{m}$) clusters after 6 hours, incorporating 60% of the beads within the field of view. The time evolution of cluster size distribution at this microscopic length scale is reproducible (Figure 2B [↗](#)).

We observed bead collection in these microscopic timelapse experiments via at least two specific motility behaviours. First, beads were directly transported upon becoming stuck to filaments, or bundles of filaments, performing ‘normal’ back-and-forth gliding motility (Figure 2C [↗](#)). Secondly, buckling and plectoneme formation in a filament were observed to transport and collect beads (Movie S5 [↗](#), S6). Buckling can occur when a gliding filament de-coordinates during a reversal. Rotation-translation coupling can then lead the filament to form a multiply twisted loop called a plectoneme [29]. We frequently observed that filament clusters (with or without beads) contained filaments that were longer than the population average, and were buckled, sometimes forming loops or plectonemes (Figure 2E [↗](#)). Beads encountered by unbuckled gliding filaments were not always transported, but when transported beads encountered another bead cluster, they generally stayed together. A fluorescent lectin, which is known to stain filamentous-cyanobacteria-associated slime [14, 29, 30], stains these bead aggregates (Figure 2D [↗](#)). These observations suggest that slime-mediated filament motility, combined with buckling, and plectoneme formation, drives particle collection and cohesion of bead clusters.

Filament physical properties determine buckling, plectoneme formation and entanglement

To test the hypothesis that buckling and plectoneme formation play an important role in particle collection, we sought a better understanding of how these relate to the motility and physical properties of the filaments. We developed a dynamical 3D model for slender filaments that move through local active gliding forces and torques. In this model, each filament is characterised by its centreline position and local orientation, and the filament can bend and twist as it glides on the surface to which it adheres, with its dynamics set by the surrounding viscous fluid (see Figure 3A [↗](#), *Methods* and *Supporting Information, SI*). The essential novelty of this model is that the filament itself is active and out of equilibrium, and additionally, the forces and torques are applied locally along its centreline, and not at its extremities as in previous steady-state mechanical studies of elastic, twistable filaments such as DNA [31–33] (see *Methods* and *SI*).

We used measured gliding speeds for *F. draycotensis* and bending constants from other motile filamentous cyanobacteria [34, 35] to derive physiologically relevant parameters (Table S1–2 [↗](#)), which we then used to simulate buckling and plectoneme formation. These events occur when the leading end of the gliding filament either ‘de-coordinates’ to reverse independently of the trailing end or becomes pinned. The two ends of the filament then produce opposing forces and torques, translating and rotating towards each other (Figure 3A [↗](#)) [29]. For the chosen parameters, the model captures both buckling and plectoneme formation as observed experimentally (Figure 3B, C [↗](#) and Movie S7 [↗](#)). The length and flexibility K of the filaments determine the type of instability, whilst adhesion and the ratio of twisting to bending modulus (G) also control the dynamics and buckled shape (see Figure S3 [↗](#) and Movie S8 [↗](#)). We derived analytical expressions for the instability thresholds for buckling and twisting instabilities, allowing us to determine critical filament lengths for bending and plectoneme formation of around $120\ \mu\text{m}$ and $320\ \mu\text{m}$ respectively (see *Methods* and *SI*). Both thresholds align well with experimental observations (Figure 3D [↗](#)). The analytical length threshold for twisting instability is slightly lower than the shortest filaments we observed forming plectonemes, which could be due to some twisting instabilities not leading to full plectoneme formation.

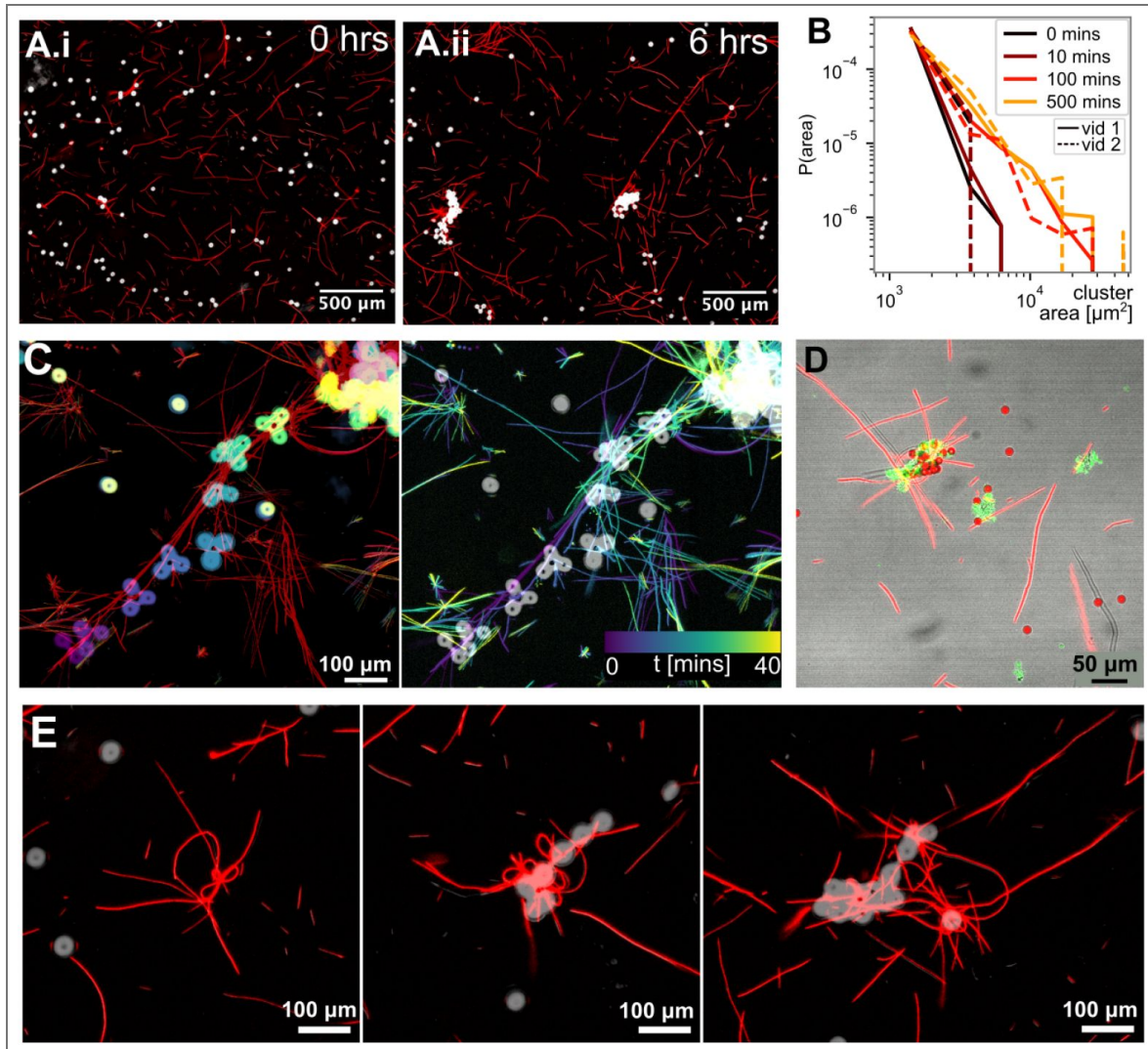


Fig. 2. Collection of beads at the microscale.

(A) Still images of *F. draycotensis* filaments (red) and polystyrene beads (white) at (i) the beginning and (ii) end of a 6-hour video (see Movie S5 [\[link\]](#)). (B) The cluster size distribution for the polystyrene beads as a function of time (shown in different colours), from two independent time-lapse experiments (dashed or solid lines). (C) Time-projected composite images showing a bundle of filaments and beads transported by their activity. Time is false-coloured from purple (0min) to yellow (40min) and is applied to the beads (left) or filaments (right). (D) Example image showing EPS concentrated within a bead cluster. EPS is stained with fluorescently labelled lectin (green), filaments are light red and beads are dark red. (E) Three different examples of entangled filament-filament (left) or filament-bead (middle, right) clusters, featuring long, looped filaments.

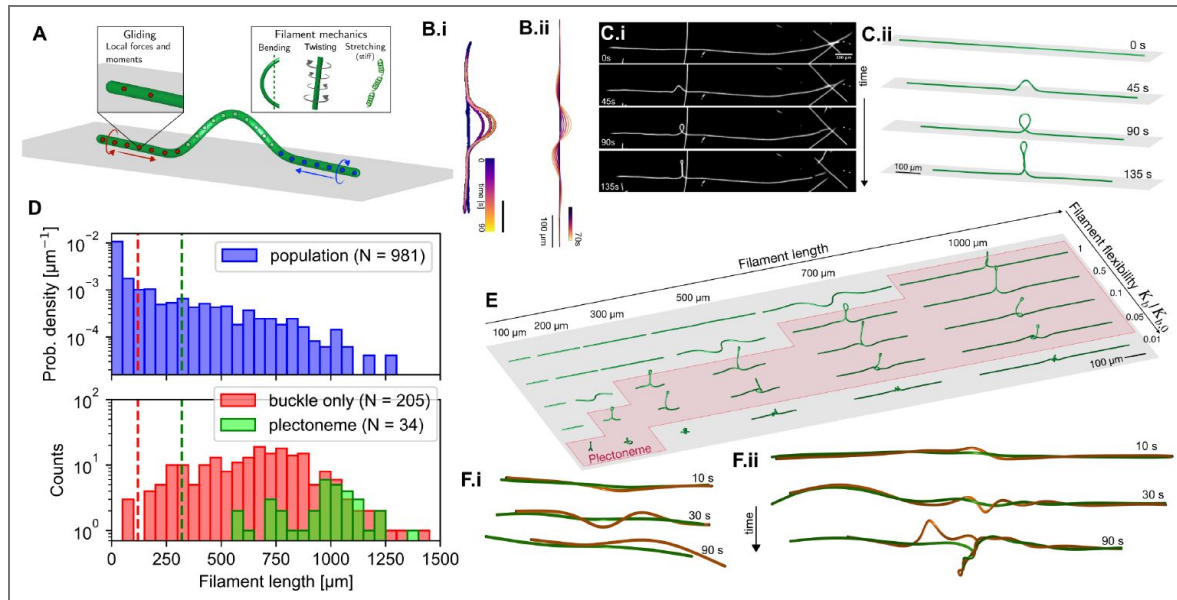


Fig. 3. A physical model of cyanobacteria as an active elastic filament.

(A) Sketch of a simulated cyanobacterial filament. Gliding on the surface is encoded by local forces and moments distributed along the filament length. The filament can bend, twist, and (minimally) stretch, with friction due to surrounding viscous fluid. The coupled rotational and translational directions of the forces are prescribed for a de-coordinated reversal. **(B)** Time-composite images of (i) a buckling *F. draycotensis* filament and (ii) an exemplar simulated filament. False-colour shows time as indicated. Scale bars 100 μm . **(C)** Images of (i) a *F. draycotensis* filament and (ii) a simulated filament forming a plectoneme (see also Movie S7). **(D)** Experimental observations of the length distributions of a filament population (blue), and instances of buckling (red) and plectoneme-forming (green) filaments within the population. Dashed vertical lines indicate thresholds for buckling (red) and twisting (green) instabilities predicted by the analytical model (see Methods and SI). **(E)** A 'phase space' for plectoneme formation. Only certain values of length and stiffness (K) result in well-behaved plectonemes, as opposed to straight, buckled, or knotted filaments (see also Figure S3 and Movie S8). **(F)** Time series of images from model simulations of pairs of flexible filaments ($K=0.1$), gliding on each other with the orange filament undergoing de-coordinated reversal. (i) Short filaments (200 μm) separate and (ii) long ones (500 μm) tangle via plectoneme formation (see also Movie 9 and Movies 10-12 for simulated and experimental entanglements, respectively).

Model simulations enable us to explore a wide range of parameters for gliding active filaments, resulting in a wide range of instability and deformation behaviours, including: absence of buckling/plectonemes; extensive buckling but no plectonemes; plectonemes; and multiple buckling/loops (see Figures [Figure 3E](#), [S3](#), and [Movie S8](#)). We found that there exists a limited region of the parameter space, relating to length and K , and to a lesser extent Γ (see *Methods* and *S1*), in which the experimentally observed single buckling and plectoneme formation occurs (Figures [3E](#) and [S3](#)).

Given the observed filament entanglements around collected beads and also the formation of macrostructures in the liquid phase in gently shaken cultures (Figure [S4D](#) and *Methods*), we also simulated the interaction of two touching filaments, gliding on each other in the absence of a surface. Within the plectoneme-forming parameter region identified above, we found filaments to readily entangle and remain entangled when a de-coordinated reversal occurs, (Figure [3F.i](#) and [Movie S9](#)). However, when the interacting filaments are too stiff or short to form plectonemes, then they do not entangle, and only glide past one another (Figure [3F.ii](#) and [Movie S9](#)). These simulations qualitatively capture observed entanglement dynamics, and in particular the involvement of plectonemes in initiation of entanglement, which is seen experimentally ([Movie S10-12](#)). Taken together, these model-based results show that buckling and plectoneme formation, as well as the associated entanglement, are confined to a limited parameter space, requiring filaments with certain physical properties.

Shorter gliding filaments cannot collect particles nor form granule macrostructures

It is difficult to test all model predictions experimentally. However, filament length is one crucial parameter affecting buckling and plectoneme formation that can be explored. To do so, we took the *F. draycotensis* cultures through a homogenization and filtering protocol (see *Methods*) to create a ‘shortened’ and ‘lengthened’ population of filaments, with the mean lengths of 40 μm and 60 μm respectively (Figure [4A](#)). We then quantified macrostructure formation in these two populations. The shortened 7 population showed gliding motility with similar characteristics to the lengthened population (Figure [4A](#)), but they did not perform particle collection or macrostructure formation over the time window in which the longer population did (Figure [1A](#) and Figure [4B, C](#)).

To further test the role of length in particle collection, we quantified the motility and macrostructure formation in an additional species of filamentous cyanobacteria: *Pseudanabaena* sp., which is taxonomically divergent from *F. draycotensis* (see *Methods* and Figure [S5](#)), but shows gliding motility with slightly faster speeds than *F. draycotensis* and similar back-and-forth movement ([Movie S13](#)). However, they are considerably shorter than *F. draycotensis*, with a mean filament length of 20 μm (Figure [4A](#)). They are therefore predicted by the physical model to be unable to form plectonemes, and indeed, we observed no such instabilities in timelapse videos of *Pseudanabaena* motility, taken over comparable timescales to *F. draycotensis* data in which plectonemes were observed (compare [Movies S1](#) and [S13](#)). Using our macroscopic assay of filament behaviour, we observed no particle collection nor aggregate formation by the *Pseudanabaena* filaments (Figure [4D](#)). Multiple days of culture growth under static conditions produced biofilm structures instead of granular macrostructures (Figure [S6](#)).

We conclude that, while motile filaments do generally form macrostructures, the nature of these structures depends on their physical properties. Noting that *F. draycotensis* and *Pseudanabaena* sp. are separated by two billion years of evolution [36] (Figure [S5](#)), such primarily physical characterisation may be more significant than other physiological variations in determining the observed diversity of macrostructures formed by different species of filamentous cyanobacteria.

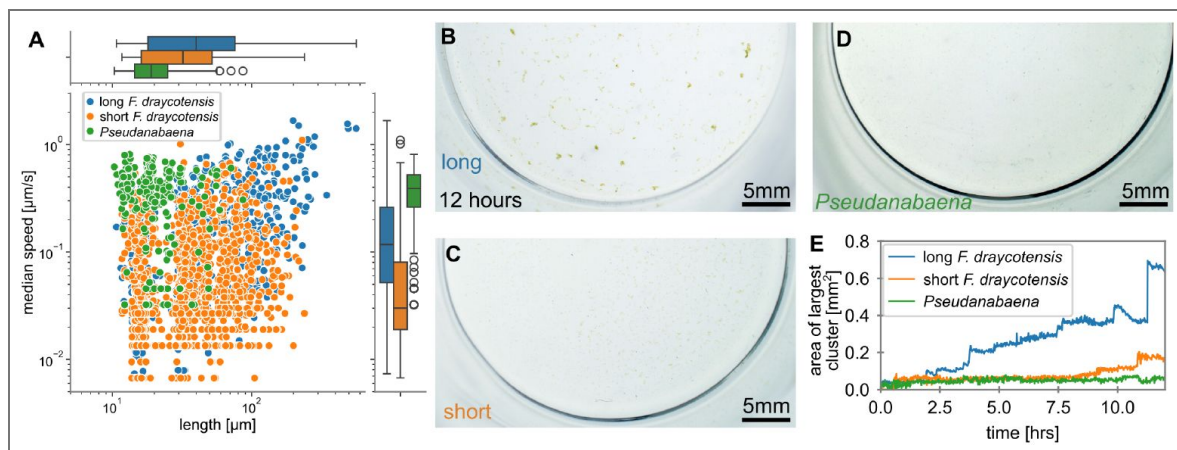


Fig. 4. Shorter cyanobacterial filaments do not collect particles.

(A) Gliding speed and filament length data for three populations of filaments: long (blue) and shortened (orange) *F. draycotensis*, and *Pseudanabaena* (green) (see Movie S13). (B–D) Images of the endpoint (12 hours after initiation) of the macro-beaker collection/aggregation assay for the same three populations. (E) Time evolution of the largest particle area visible in the beakers for the three populations, coloured as in (A).

Discussion

Here we presented and characterised a surprising, emergent behaviour in motile filamentous cyanobacteria: active collection of solid particles from the environment and incorporation of these into persistent macroscopic granular structures. In particular, our results show that the ability of filaments to buckle and form plectonemes promotes filament entanglement and particle collection, thereby leading to the construction of granular macrostructures. The capacity for buckling and plectoneme formation is unlikely to be confined to a single clade; rather, it should arise in any species whose filaments satisfy critical length and flexibility criteria derived here. This insight, as well as the presented modelling and experimental approaches and model systems, will allow development of a predictive framework for understanding the diversity of macrostructures observed in different natural and biotechnological settings and in different filamentous (cyano)bacterial species.

While we described particle collection and macrostructure formation under laboratory conditions, the presented model systems will help us to study these processes under more specific environmental and biological scenarios, such as in the water column or under different flow regimes on surfaces. For example, the *F. draycotensis* cultures studied here are obtained from a freshwater reservoir, where granular aggregates are commonly observed in the water column, rock pools and on shore rocks (Figure S4 [\[4\]](#)). In the laboratory, we observed *F. draycotensis* macrostructure formation in the bulk liquid phase of gently shaken cultures (Figure S4 [\[4\]](#)). However, further systematic studies are needed to quantify granule formation in other settings, such as varied flow conditions or low filament densities. The presented findings and their future extensions will be useful in biotechnology, to help with bioengineering of cyanobacterial granules for water remediation applications [\[2, 5\]](#) using targeted, suitably long and flexible filamentous species and culturing conditions.

These results call for studies of macrostructure formation in conjunction with stronger emphasis on quantification of filament physical properties. Notably, the presented physical model predictions, regarding the filament length and stiffness required for plectoneme formation and particle collection, were sufficient to explain the difference in collection behaviour in species of two deeply divergent cyanobacterial orders, without invoking lineage-specific biochemistry. Filamentous multicellularity and gliding motility are ancestral traits broadly conserved across cyanobacteria [\[16, 27\]](#), suggesting that these biophysical principles may extend to other filamentous cyanobacteria found in environmental mats, biofilms, and stromatolites. Plectoneme formation and macrostructure formation may also be present in other filamentous prokaryotes, such as the sulfide-oxidising cable bacteria (*Desulfobulbaceae*), whose long and flexible filaments form dense sedimentary networks that affect local pH and mineral precipitation profiles [\[53\]](#). This could affect local concentration of reactive mineral phases, with important consequences for sediment geochemistry.

It is notable that we first observed particle collection with iron-containing precipitates. Iron is a common limiting nutrient for cyanobacteria in many environments where it readily precipitates out of solution. Whilst we found particle collection to be a general capability, extending to any inert solid particle, it is plausible that micron-sized particles that filaments encounter in nature would include mineral precipitates and nutrient-rich organic matter. Collection of these into macrostructures can be interlinked with embedding of, or subsequent colonisation by, other bacteria. Indeed, cyanobacterial macrostructures are repeatedly shown to be key habitats for the emergence of interlinked microbial metabolisms. Such microbial interactions and overall microbial diversity in granules is shown to be disrupted when macrostructure formation is inhibited in *F. draycotensis* communities [\[4\]](#). Thus, particle collection and macrostructure formation can have significant downstream functional consequences.

We find that motility-driven macrostructure formation is a highly dynamic process, involving emergent particle collection. Not only is it driven by motile filaments, but the resulting macrostructures can also move, disassemble, and reassemble. These emergent dynamics result from the interaction of thousands of filaments: such population-level behaviours are widely

studied in active matter physics. Such studies have found phase transitions in the patterning of certain filamentous cyanobacteria due to densitydependent nematic ordering [54]; and photoresponsive morphologies in multi-filament bundles [28]. The co-development of models and experiments in the readily-culturable filamentous species like *F. draycotensis* will enable further study of these dynamic, emergent complex behaviours in filamentous active matter.

Methods

Culturing of cyanobacterial community

All cyanobacterial cultures are established from a freshwater source (Draycote Reservoir, UK) and are maintained under stable laboratory conditions as described before [4]. Cultures are kept in the laboratory under regular serial sub-culturing in BG11+ medium (DSMZ medium reference number 1593), with additional vitamins [4]. Subculturing was done approximately monthly, with cultures grown under continuous 12 h/12 h light/dark cycles with white, fluorescent illumination of 14-20 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ at room temperature under static conditions. For each passaging step, we performed a 1:50 dilution by transferring 150 μL of re-suspended filamentous culture into a final volume of 30 mL of BG11+.

Granule sectioning

Macrostructures of different ‘age’ and size were removed from cultures using a widebore pipette tip to avoid deformation. Structures were fixed using a glutaraldehydelysine fixation process, adapted from a method for preserving bacterial capsules [37]. Firstly, photogranules were placed in PBS for 5 minutes to remove amines. The granules were then transferred between a series of solutions, all prepared in PBS at room temperature: 50 mM lysine (10 minutes), then 50 mM lysine and 2.5% glutaraldehyde (10 minutes), then 2.5% glutaraldehyde (2 hours), finally followed with 3 transfers into PBS (5 minutes each). Photogranules are snap frozen within a PVP:HMPC hydrogel matrix (2.5% PVP and 7.5% HMPC, g/g) for cryosectioning, based on methods used in [38]. A Bright OTF7000 Cryostat was used to perform cryo-sectioning (chamber: -20°C , sample head: -14°C). Samples were sectioned to a thickness of $100\mu\text{m}$, and collected onto cover glass slides for imaging.

Iron analysis of granule cores

A *F. draycotensis* culture was allowed to grow under normal growth conditions described above and regularly checked for visible yellow particles over two days. Emerging, visible particles are collected by pipette under sterile conditions and placed into 11 collection tubes containing high-purity water. These are centrifuged ($10,000 \times g$, 10 minutes) to pellet the yellow material, the water is removed and the granules resuspended into 70% ethanol solution for 30 minutes to terminate cell growth. The collection tubes are centrifuged again, and washed three times with 70% ethanol. The material is then centrifuged, and resuspended again in ultra-pure water. The water wash step is repeated twice to remove ethanol. The sample is then placed into a vacuum condenser (Genevac) to remove the water.

X-ray Photoelectron Spectroscopy (XPS) of granule cores

XPS was performed by attaching a sample of the extracted granule core to electrically-conductive carbon tape, mounted onto a sample bar with a layer of filter paper between the samples and the sample bar to ensure electrical isolation and hence mitigate differential charging, before being loaded into a Kratos Axis Ultra DLD spectrometer which possesses a base pressure below 1×10^{-10} mbar. XPS measurements were performed in the main analysis chamber, with the sample being illuminated using a monochromated Al K α x-ray source ($h\nu = 1486.7 \text{ eV}$). The measurements were conducted at room temperature and at a take-off angle of 90° with respect to the surface parallel. The core level spectra were recorded using a pass energy of 20 eV (resolution approx. 0.4 eV), from an analysis area of 300×700 microns. The work function and binding energy scale of the spectrometer were calibrated using the Fermi edge and 3d $5/2$ peak recorded from a polycrystalline Ag sample prior to the commencement of the experiments. To prevent surface

charging the surface was flooded with a beam of low-energy electrons from a charge neutraliser throughout the experiment and this necessitated recalibration of the binding energy scale. To achieve this, the C-C/C-H component of the C 1s spectrum was referenced to 285.0 eV. The data were analysed in the CasaXPS package using Shirley backgrounds and mixed Gaussian-Lorentzian (Voigt) lineshapes. For compositional analysis, the analyser transmission function has been determined using clean metallic foils to determine the detection efficiency across the full binding energy range.

Inductively coupled plasma mass spectrometry (ICP-MS) analysis of granule cores

Cultures were fractioned into supernatant, cell pellet and yellow granular core, as explained above. Samples of BG11+ media was also included for analysis, to serve as a media reference. All sample fractions were placed in the Genevac and dried until total dryness was achieved. Dried sample was then dissolved in 70% high-purity nitric acid, diluted into water (1:20) and then filtered through 0.22 μ m PES filters, and analysed on the ICP-MS. A SPS4 auto-sampler Agilent 7900 was used for ICP-MS. The following equipment conditions were set: spray chamber temperature: 15°C, plasma gas flow rate: 15 L min⁻¹, carrier gas flow rate: 0.9 L min⁻¹, nebuliser gas flow rate: 1.05 L min⁻¹, auxiliary gas flow rate: 0.9 L min⁻¹, and helium mode gas flow rate: 4.0 mL min⁻¹. The forward power was 1550 watts, with a flow rate of 0.1 rps. Detector analog HV 2152V, pulse HV 956V, with a discriminator voltage of 5 mV and a 4.3 mL helium gas flow rate. Prior to the sequence, the instrument was auto-tuned, and during the analysis, the P/A was modified using calibration standards.

Shaken cultures

For the ‘swirling’ flask shown in [Figure S4D](#), cultures were photographed after 16 days of growth. The culture was grown in an Algem 1.4 (ALGae Environment Modelling lab scale photobioreactor), by Algenuity, at a rotation rate of 80 rpm. The flask was illuminated from beneath with LEDs at $\approx 60 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, on a 12hr/12hr light/dark cycle. The culture was set up using a 1:50 dilution (8ml culture in total 400 ml BG11+ media).

Macroscopic aggregate formation experiments

The process of aggregate formation was recorded on a macroscopic scale in 50mL glass beakers of approximately 37 mm diameter. The beakers were inoculated with 0.4 mL of a mature culture sample, aged between 35 and 70 days, and diluted 1 in 50 into 19.6 mL of fresh BG11+ media to give a total fluid volume of 20 mL. As the microbial communities found in culture flasks of this age have usually formed persistent, cohesive granules, we homogenise the inoculant to break up these structures before setting up the beaker experiments. We do this by pipetting approximately 1 mL of sample up and down in an Eppendorf tube. We then dilute the homogenised inoculant to an optical density of 0.4 compared to a media blank (absorbance at 750 nm, measured using a SPECTRONIC 200 Spectrophotometer, Thermo Fisher Scientific).

The experiment is set up by placing the beakers containing only the fresh media under a DSLR camera (Nikon) held on a post at sufficient distance to allow the entire base of the beaker to be focused in the imaging plane. The beakers are lit from below using a sheet of LED lights, diffused through multiple layers of white diffusion paper (Lee Filters, 216 White Diffusion Lighting Gel) to give a brightness of $36 \mu\text{mol photons m}^{-2} \text{s}^{-1}$ at the base of the beakers. The camera is focused by hand at the beginning of the experiment, with adjustments made during the imaging time as necessary. Once the initial focus is set, the sample is added to the media in the beaker, and the beaker is covered with a plastic petri dish lid to reduce evaporation. Images are then taken once every minute for up to 2000 minutes (33.3 hours).

The macroscopic collection assay in [Figure S2](#) was performed in the same glass beakers as all other ‘macroscopic aggregate formation’ experiments. For [Figure S2A](#), we used higher initial densities of culture, and rotated the beakers gently at 150rpm (using a Heidolph Unimax 1010

rotary shaker) to physically concentrate any solids to the centre of the beaker via fluid flow. This allowed easy visibility of either active collection (in the case of the motile culture) or precipitation (for the other two cases) after 5 hours. Additional agitation was then performed by manually swirling and tapping the beaker. For Figure S2B [\[30\]](#), we used a derived bacterial community lacking *F. draycotensis*, grown on BG11+ with added 0.8% ($\approx 44\text{mM}$) glucose. As this ‘no-cyanobacteria’ community soon becomes optically dense, black polystyrene beads (Thermo Scientific Black ChromoSpheres Polymer Microspheres, $284\mu\text{m}$ diameter) are added to the beakers to allow particle collection to be easily visualised.

Macroscopic aggregate formation image analysis

Images of macroscopic aggregate formation within glass beakers are segmented and analysed to measure particle size and speed. Segmentation is performed on the blue channel of the RGB images, which has the largest contrast between the background and the aggregates. The images are automatically thresholded to segment the particles from the background, then particle positions and areas are measured using the Analyze Particles function in ImageJ. The particles are then tracked using the Trackpy module in Python. This code is available at https://github.com/OSS-Lab/Cyanobacteria_particle_collation [\[31\]](#).

Microscopy imaging of filaments and beads

Particle collection at the microscale was imaged using an Olympus IX83 inverted microscope with a Photometrics CoolSnap-HQ2 camera. Time-lapse videos were taken in brightfield and fluorescence imaging. Chlorophyll autofluorescence of the filaments was imaged using a CoolLED pE-300 illumination system and a Chroma HcRed1 filter cube (Excitation filter: $\approx 550\text{--}600\text{ nm}$, Dichroic mirror: Q610lp, Emission filter: HQ640/50m, $\approx 615\text{--}665\text{ nm}$). We used time intervals of 10s–60s and an exposure time of 100ms. Exemplar timelapse images are available at <https://zenodo.org/uploads/18680430> [\[32\]](#).

Experiments were performed in a 6 cm glass petri dish base, with a plastic petri dish used as a lid. 12mL of BG11+ media was added to the dish, along with a small quantity of $48\mu\text{m}$ diameter black polystyrene microspheres (ChromoSphere Dry Dyed Polymer Particles, Black, BK050). Between 0.2–0.4 mL of homogenised mature sample of the cyanobacteria was then added and imaging was initiated immediately afterwards.

Lectin staining

Following one of microscale collection experiments described above, we stained the sample with a fluorescein-tagged lectin stain (RCA-120, Vectorlabs.) This lectin agglutinates sugars including galactose [\[30\]](#), which is known to be a major component of the EPS produced by *F. draycotensis* [\[4\]](#). Images of the stained sample were taken using a Zeiss 880 confocal microscope. The RCA-120 lectin was imaged using a 488 nm argon laser, with a PMT emission range of 510–550 nm (green channel), whilst the autofluorescence from the filaments and beads was imaged using a 633 nm HeNe and an emission range on the PMT of 638–700 nm. Imaging was performed with a pinhole diameter of $45.96\mu\text{m}$, and a MBD 488/561/633 beamsplitter was used to allow imaging across both channels simultaneously.

Creation of shortened *F. draycotensis* populations

Populations of *F. draycotensis* with shortened filaments were created by homogenising $\approx 1.5\text{ mL}$ mature culture, then allowing it to settle for ≈ 30 minutes in a 2 mL Eppendorf tube, resulting in a loose aggregate. The ‘supernatant’ of this aggregate is extracted and filtered through a $40\mu\text{m}$ mesh (Greiner Bio-One EASYstrainer Cell Strainer, Small) to give the ‘short population’. The loosely aggregated filaments are resuspended and homogenised by pipetting up and down in order to give the ‘long’ population. The densities of the two populations are matched by optical density measurements, and the length distribution and motility of both of the populations are confirmed by microscopy.

Length and speed distributions of filament populations

We used time-lapse microscopy images of filaments gliding on a glass surface to measure length and speed distributions of various populations of filaments. Time-lapse images are available at <https://zenodo.org/uploads/18680430>. Samples were prepared on glass slides with an imaging spacer (Thermo Fisher 25 μ L Gene frame) and sealed with a plastic cover slip, and imaged on an Olympus IX83 inverted microscope in the HcRed fluorescence channel. The filament density was sufficiently high that any frame of the video contains a significant fraction of filaments that overlap with one another. We therefore implemented an automated segmentation routine to extract the filaments, based on the method outlined in [39]. This code is available at https://github.com/OSS-Lab/Cyanobacteria_particle_collation. The filament trajectories were then linked in Python using the Trackpy module in order to calculate filament speeds.

Phylogenomic analysis

A phylogenomic tree was constructed from 29 cyanobacterial genomes spanning six orders (Gloeobacterales, Synechococcales, Pseudanabaenales, Cyanobacteriales, Phormidesmiales PCC-6307, Leptolyngbyales, and Elainellales). Reference genomes were obtained from NCBI, preferentially selecting RefSeq assemblies where available (see Table S3). Additional genomes were derived from metagenomic analysis of cyanobacterial enrichment cultures from a recent study [40]. Six ribosomal protein markers (Ribosomal L1, L2, L3, L4, L5, and L6) were identified from each genome using Hidden Markov Model (HMM) profiles from the Bacteria71 single-copy core gene collection in anvi'o [41]. For each genome, the best-scoring hit per marker was retained, and amino acid sequences were extracted and concatenated, yielding an alignment of 1,472 positions. Poorly aligned regions were removed using trimAl v1.4 [42] with a gap threshold of 0.5, retaining columns in which at least 50 of sequences contained a residue, resulting in a trimmed alignment of 1,322 positions. Maximum-likelihood phylogeny was inferred using FastTree v2.2.0 [43] with the JTT-CAT substitution model and 20 rate categories. Branch support was assessed using Shimodaira–Hasegawa-like approximate likelihood ratio test values computed from 1,000 resamples as recommended [44]. The tree was rooted on *Gloeobacter spp.*, which represents the earliest-diverging cyanobacterial lineage [36]. Morphological traits (filamentous or unicellular) and habitat information (freshwater, terrestrial, marine, or symbiont) were assigned to each strain based on published literature. The tree was visualised and annotated using TreeViewer [45].

Filament modelling and simulation methods

Filament model

We developed a 3D physical model of slender, elastic filaments undergoing active motion to describe the gliding motility, buckling, and plectoneme formation of the filamentous cyanobacteria. The model code is available at https://github.com/OSS-Lab/Cyanobacteria_particle_collation. This model builds on an existing framework that combines the classical equilibrium theory for passive elastic filaments—formulated through Kirchhoff rod models for systems such as DNA [31–33]—with immersed-boundary methods for describing filament dynamics in a viscous fluid [46–49]. We extend this approach to include the local active forces generated by gliding motility, and interactions with the external environment. We approximate the medium in which the cyanobacteria move as a viscous fluid above a solid surface, which causes friction and sets the dynamics of filament motion. Within this formalism, any filament deviating from its relaxed, straight configuration experiences elastic forces and torques that oppose bending and twisting, and these are transmitted to the viscous fluid as hydrodynamic singularities. The filament then moves and reorients according to the local active flow field [46–49].

Filament description

The configuration of the filament can be described by the centreline position $\mathbf{X}(s)$ of the filament at a given point s and its local orientation, given by the local material frame, an orthonormal director basis $[\mathbf{d}_1(s), \mathbf{d}_2(s), \mathbf{d}_3(s)]$. In the classical Kirchhoff framework, $\mathbf{d}_3(s)$ is the tangent to the

curve $\mathbf{X}(s)$, and $\mathbf{d}_1$ and $\mathbf{d}_2$ point normal to the curve $\mathbf{X}(s)$. The twist vector $\boldsymbol{\kappa}(s)$ is then defined from the derivatives of the director basis with respect to the arclength, $\partial_s \mathbf{d}_i = \boldsymbol{\kappa} \times \mathbf{d}_i$, where $\boldsymbol{\kappa} = \kappa_1 \mathbf{d}_1 + \kappa_2 \mathbf{d}_2 + \Omega \mathbf{d}_3$. The elements of the twist vector are known as filament curvatures, and the curvature in the third direction Ω is referred to as the twist density. In our dynamic simulations, because the filament moves with the local flow, as in [47], we allow the material frame to deviate from the tangent and normal vector triad as it rotates with the flow. We also allow for some weak extension of the filament, but impose a large stretch modulus b_3 and shear force constants b_1 and b_2 .

Bending and twisting forces and moments

With these formulations in place, we can write down the equation of state describing the elastic forces ($\mathbf{F}$) and torques (i.e., moments, $\mathbf{M}$) on a point s on the filament at time t for a given filament configuration as;

$$\begin{aligned} \mathbf{M}(s, t) &= K_b (\kappa_1 \mathbf{d}_1 + \kappa_2 \mathbf{d}_2) + K_t \Omega \mathbf{d}_3, \\ \mathbf{F}(s, t) &= b_1 \mathbf{d}_1 \cdot \partial_s \mathbf{X} + b_2 \mathbf{d}_2 \cdot \partial_s \mathbf{X} + b_3 (\mathbf{d}_3 \cdot \partial_s \mathbf{X} - 1). \end{aligned} \quad (1)$$

where we introduced the twist coefficient, K_t , and the bending coefficient, K_b , describing the resistance of the filament to twisting and bending, respectively, and b_3 and b_1, b_2 describe its resistance to stretch and shear, with $b_i > K_b, K_t$.

To characterise the filament mechanics, we introduce the relative bending stiffness $K = K_b/K_{B,0}$ relative to the mean experimental value from several cyanobacteria filaments $K_{B,0}$ [34, 35], and the ratio of twist-to-bending moduli, $\Gamma = K_t/K_b$.

Active gliding

To model active gliding motility, we incorporate the active forces and torques from each cell of the filament. While the exact underlying mechanism for the gliding is not well understood, we know that - in *F. draycotensis* filaments - each cell translates and rotates as it moves [29]. As a minimal model for gliding, we consider that at each point, the filament experiences local forces and moments along the local tangent vector, with densities $\mathbf{f}_{\text{glide}}$ and $\mathbf{n}_{\text{glide}}$, respectively. We impose that the gliding only occurs when the filament is in contact with a solid surface, which can be another filament.

Surface interaction

We also impose a steric repulsion between the surface and the filament. The steric interactions are active where the centreline of the filament is touching the surface, $x < 0$, are directed normal to the surface, and increase linearly with x , $\mathbf{f}_{\text{ste}}(s, t) = c_{\text{rep}} |x(s, t)| \delta_{x < 0} \hat{\mathbf{e}}_x$.

Finally, we consider adhesion between the surface and the filament, with a typical strength c_{adh} , the effect of which we explore in simulations. It is active on portions of the filament that are in the adhesion region $x \in [\delta_{\text{adh}}/2, \delta_{\text{adh}}]$, with $\mathbf{f}_{\text{adh}} = \frac{c_{\text{adh}}}{2} \left[1 + \cos \left(2\pi \left(\frac{x}{\delta_{\text{adh}}} - \frac{3}{4} \right) \right) \right] \hat{\mathbf{e}}_x$, so that the force is oriented perpendicular to the surface, cancels at the bounds of the adhesion layer, and reaches a maximum c_{adh} in the middle of the filament.

Force balance

With the above definitions in place, we can write the force balance on a point s on the filament at time t . To do so, we use a coupled system of equations where the elastic forces and torques generated by the bending and twisting of the filament are transmitted to the viscous fluid. The rod experiences the elastic forces and torques from eq. (1), active force and torque densities $\mathbf{f}_{\text{glide}}$, $\mathbf{n}_{\text{glide}}$ from gliding, viscous forces and torques from the fluid $\mathbf{f}_v$, $\mathbf{n}_v$, and steric interactions with the surface $\mathbf{f}_{\text{ste}}$. The force and moment balance is;

$$\begin{aligned} \partial_s \mathbf{F}(s, t) + \mathbf{f}_v(s, t) + \mathbf{f}_{\text{glide}}(s, t) + \mathbf{f}_{\text{adh}}(s, t) + \mathbf{f}_{\text{ste}}(s, t) &= \mathbf{0}, \\ \partial_s \mathbf{M}(s, t) + \mathbf{d}_3 \times \mathbf{F}(s, t) + \mathbf{n}_v(s, t) + \mathbf{n}_{\text{glide}}(s, t) &= \mathbf{0}. \end{aligned} \quad (2)$$

Surrounding fluid

Our simulations for the elastic rod in a viscous fluid relies on the immersed boundary method for a slender filament [46–49], which describes the interaction between the active elastic filaments and the surrounding viscous fluid. In this setup, the elastic and gliding forces and torques are directly transmitted to the fluid according to eq. (2) [46], and they generate an active flow around the filament. The filament in turn moves and deforms with the local fluid velocity.

The surrounding viscous fluid verifies the incompressible Stokes equations for the fluid velocity $\mathbf{u}$ and pressure $\mathbf{p}$,

$$-\nabla p(\mathbf{x}, t) + \mu \Delta \mathbf{u}(\mathbf{x}, t) + \mathbf{g}(\mathbf{x}, t) = \mathbf{0}, \quad \nabla \cdot \mathbf{u}(\mathbf{x}, t) = 0, \quad (3)$$

with $\mathbf{g}$ denoting the distribution of forces per unit volume imposed on the fluid.

As the propulsion, elastic, and adhesion forces on the filament are all transmitted to the fluid as point forces and torques, they generate a local force per unit volume distribution, $\mathbf{g}(\mathbf{x}, t)$. To avoid singularities in the flow velocity, we use a regularisation kernel ψ_ϵ such that the forces are not applied on singular points of the filament but over an area of typical size ϵ .

$$\begin{aligned} \mathbf{g}(\mathbf{x}, t) = & \int (-\mathbf{f}_v(s, t)) \psi_\epsilon(\mathbf{x} - \mathbf{X}(s, t)) ds + \\ & \frac{1}{2} \nabla \times \int (-\mathbf{n}_v(s, t)) \psi_\epsilon(\mathbf{x} - \mathbf{X}(s, t)) ds \end{aligned} \quad (4)$$

With $\psi_\epsilon(r) = \frac{15\epsilon^4}{8\pi(r^2 + \epsilon^2)^{7/2}}$. We then compute the local velocity of the fluid given this distribution of local forces according to eq. (3) [46] as the sum of regularised Stokeslet (left integral) and rotlet (right integral) singularities.

In addition, we enforce the no-slip boundary condition at the solid surface $\mathbf{u}((-\delta, y, z), t) = \mathbf{0}$ by using regularised image singularities for the Stokeslets and rotlets [50–52]. That is, we ensure that the fluid at the solid surface has zero velocity by adding an adequate combination of local flow singularities at the mirror points from the active filament, with their exact expression given in [52].

Filament dynamics

The filament then moves and rotates with the local fluid velocity, as

$$\partial_t \mathbf{X} = \mathbf{u}(\mathbf{X}, t), \quad \partial_t \mathbf{d}_i = \mathbf{w} \times \mathbf{d}_i, \quad (5)$$

with $\mathbf{n} \times \Delta \frac{\tau}{\Gamma} = \mathbf{M}$ the local flow vorticity.

The numerical implementation for the resulting system follows [47]. We discretise the filament in sections of size ds . To compute the derivatives of the forces and moments along the arclength, we compute them at midpoints of the filament and use auxiliary reference frames at these midpoints. We then use a forward Euler algorithm to solve the coupled system of equations. In all our simulations, we use parameter values from the literature on filamentous bacteria when available [29, 34, 35], and infer physically consistent values for the other parameters. The parameters used in the simulations are given in Table S1 [46] for physical properties and in Table S2 [46] for numerical implementation, and their influence on simulation outcomes are discussed in the SI.

Two filament interactions

We extended these simulations to explore the buckling and twisting dynamics of two filaments moving against each other. In the two filament system, gliding along the filament direction occurs for points that are within the contact region of the other filament (instead of the surface) and generate an equal and opposite force on the other filament. We also account for steric interactions between the filaments. We study in particular their entanglement when a reversal occurs in the middle of one of the filaments, as seen experimentally. We find that plectoneme formation regime

(long, flexible filaments) in the two-filament system leads to entanglement, as depicted in Figure 3F.ii. Shorter filaments, which wouldn't form plectonemes on the surface, instead tend to glide past one another and separate even when a reversal occurs (Movie S9).

Simulation parameters

In all our simulations, we use parameter values from the literature on filamentous bacteria when available [29, 34, 35], and infer physically consistent values for the other parameters (see Table S1).

For the motility, we use the experimentally measured gliding force density $f_{\text{glide}} = 1 \text{ nN} \cdot \mu\text{m}^{-1}$ [35]. We assume that bacteria glide over a film of thickness of $1 \mu\text{m}$ above the solid surface, and set the medium viscosity μ to match the experimental gliding speed of the filament $v_{\text{glide}} \approx 1 \mu\text{m} \cdot \text{s}^{-1}$ for a $200 \mu\text{m}$ filament based on *F. draycotensis* filament speeds measured here and previously [29]. We then chose a torque density so that the rotation rate matches its measured experimental value $\omega_{\text{glide}}/v_{\text{glide}} \approx 0.87 \mu\text{m}^{-1}$ [29].

The bending of filamentous cyanobacteria has been studied experimentally [34, 35], but we couldn't find in the literature values for the twisting modulus of filamentous bacteria, and assumed a unit twist-to-bend ratio Γ , but also explored other values in simulations. We take the adhesion force to be stronger than the gliding forces, $c_{\text{adh}} = f_{\text{glide}}$. For smaller values of this coefficient, buckling leads to a detachment from the surface, contradicting experimental observations (see Figure 3E and S3). The numerical parameters are chosen for the simulations to be stable, relying on previous implementations of elasto-hydrodynamic simulations [49].

Data availability

This work generated computational models and microscopy timelapse images, as datasets. These are available at GitHub (<https://github.com/OSS-Lab/Cyanobacteria>) and a dedicated Zenodo repository (<https://doi.org/10.5281/zenodo.18680429>).

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Additional information

Author contributions

RP, AT, and OSS conceptualised and developed the methodology of the study. RP, MC, KC, and AS performed experiments and analyses, while AT developed models and ran simulations. RP, AT, and OSS performed formal analysis and visualisation of the data. RP, AT, and OSS wrote the manuscript and all authors contributed to its review and editing. OSS obtained funding for the study, and OSS and MC were responsible for project administration.

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Additional files

[Supplementary Text](#) 

[Supplementary Movie S1](#) 

[Supplementary Movie S2](#) 

[Supplementary Movie S3](#) 

[Supplementary Movie S4](#) 

[Supplementary Movie S5](#) 

[Supplementary Movie S6](#) 

[Supplementary Movie S7](#) 

[Supplementary Movie S8](#) 

[Supplementary Movie S9](#) 

[Supplementary Movie S10](#) 

[Supplementary Movie S11](#) 

[Supplementary Movie S12](#) 

[Supplementary Movie S13](#) 

Note

This reviewed preprint has been updated to correct a corresponding email address.

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Peer reviews

Reviewer #1 (Public review):

Summary:

In this manuscript, the authors investigate the mechanisms underlying macrostructure formation in a freshwater filamentous cyanobacterium strain, *F. draycotensis*, focusing on how its ability to aggregate and form these structures depends on the physical properties of the filaments. Using experimental observations, they demonstrate that the cyanobacterium actively captures and surrounds particles, a process driven primarily by gliding motility.

To explain these physical dynamics, the authors present a 3D model indicating that particle collection relies on filament length, as well as a specific mechanical response, namely, filament buckling and the subsequent formation of loops of bundles of filaments. While the authors have previously documented the buckling and looping characteristics of this strain, this study provides new insight by demonstrating that these physical phenomena are essential for particle capture and collection.

Strengths:

This manuscript benefits from a rigorous and detailed quantitative analysis of video recordings, which clearly documents the motility, buckling behaviour, and particle collection dynamics of the filaments.

The authors effectively validate their hypothesis by using a naturally shorter filamentous strain, which fails to collect particles, suggesting that filament length is indeed a critical parameter.

To further confirm the length dependency within the same species, the authors experimentally generated shorter filaments of *F. draycotensis*. The fact that these shortened filaments also lose the capacity to collect particles provides strong evidence supporting their proposed mechanism.

Weaknesses:

There is a conceptual concern. The authors linked the specific physical properties of this strain to evolutionary data, highlighting that the studied lineages diverged approximately two billion years ago. This creates a misleading impression that particle collection via flexible looping filaments is a recent evolutionary adaptation. However, particle collection has been observed in other cyanobacteria, such as *Trichodesmium*, which features short, rigid filaments. Therefore, the term "emerging" does not seem appropriate for the title and text. The capacity to collect particles in the studied strain *F. draycotensis* appears to be primarily a function of physical characteristics (filament length and flexibility) rather than evolutionary age. Any cyanobacterial strain possessing similar physical properties is likely to exhibit comparable behaviour, rendering the evolutionary timeframe largely irrelevant to the core mechanism. In addition, the phylogenetic tree presented in Figure S5 does not reflect the current consensus on cyanobacterial evolution and systematics and does not align with modern phylogenomic frameworks (see, for example, Strunecky et al., 2023 <https://doi.org/10.1111/jpy.13304>). There is also no such order Cyanobacteriales, which has been mentioned in a few older publications but is clearly outdated.

Another concern is that the authors nearly completely ignore the role of type IV pili in the gliding motility of cyanobacteria, including filamentous strains. For a long time, there was a misconception that the gliding motility of cyanobacteria was due to slime protrusion. Slime plays a role in this process. However, several studies have shown that filamentous strains also use type IV pili to glide on surfaces. The authors should discuss this and include it in their model. In addition, the authors concluded that gliding motility is responsible for particle collection by *Fluctiforma draycotensis*. Although I believe that their conclusion is correct, there might be several limitations to the experiments which allow for other reasons to be considered. Their conclusions were based on the use of a non-motile strain and an unspecified community without the motile *Fluctiforma draycotensis* strain. The problem I see here is that it is not clear why this strain is not motile; it could be because of the lack of type IV pili, mutations which alter their functionality, defects in slime secretion, any other mutation (e.g. in chemoreceptors), cellular structure, metabolism, or combinations of these. Furthermore, it is possible that the community changes its composition and behaviour when it lives without the cyanobacterium with a rich carbon source (glucose) or with a non-motile cyanobacterium which may not secrete slime or, for example, a signalling component which controls behaviour of the bacteria in the community. For that reason, the authors should be more cautious with their conclusion that solely motility behaviour of *Fluctiforma draycotensis* is responsible for particle collection. Additional factors might be responsible for these effects.

<https://doi.org/10.7554/eLife.112517.1.sa3>

Reviewer #2 (Public review):

Summary:

The authors studied aggregation, buckling, and particle collection by the filamentous cyanobacterium *Fluctiforma draycotensis*, as well as by the filamentous *Pseudanabaena* sp. (order Pseudoanabinales). They performed a range of experiments, from imaging individual gliding filaments to multiple-day experiments showing the formation of large aggregates around a particle formed from a precipitate. They also developed a model of buckling filaments to argue that the ability of elastic filaments to collect particles and form macrostructures is confined to a part of the filament phase space in terms of length and flexibility, meaning that gliding combined with certain filament length and flexibility naturally reproduces the observations.

Strengths:

This is an impressive study that uses multiple tools to connect macrostructure formation with filaments' gliding motility and buckling. It adds an important perspective on the biological and physical factors at play in the emergence of aggregates.

Weaknesses:

The authors ignore the possibility that filament behavior plays an important role in the emergence of the observed patterns. Cyanobacteria have been shown to control their gliding motility (Pfreundt et al Science 2023; Kurjahn et al Nature Comm 2024), and their molecular motors are known to be regulated by chemotaxis-like signaling pathways (Risser ARM 2025). As far as I know, how the coordination between the pulling agents along an individual filament works is actively debated, but there seems to be little doubt that it exists.

To illustrate this point better, note that the aggregation observed by the authors is consistent with the length-dependent ability of filaments to coordinate gliding (I'm not saying this is how it works in *Fluctiforma draycotensis*; I'm saying it's consistent). Suppose the coordination requires sufficiently long filaments, which could be the case when signaling molecules travel along the filament, propagating information about when individual pulling agents should reverse. In such a model, short filaments act randomly because they fail to coordinate gliding by the time they glide off nascent aggregates, whereas longer filaments can perform informed reversals because they have more time for coordination. Such behavior then explains the lack of aggregation in *Pseudanabaena* sp. (via behavior, not lack of stiffness). Note that *Trichodesmium* is stiff; its filaments do not buckle, yet *Trichodesmium* forms organized aggregates via tightly controlled motility. Note also that, as the authors report, since *Pseudanabaena* sp. is both shorter and faster, its filaments have relatively (to the time needed to glide the filaments' length) little time to coordinate reversals. In my opinion, whether the observed patterns passively emerge from gliding and buckling or result from active behavior remains an open question.

I also have a small suggestion regarding this statement on model novelty:

The essential novelty of this model is that the filament itself is active and out of equilibrium, and additionally, the forces and torques are applied locally along its centreline, and not at its extremities as in previous steady-state mechanical studies of elastic, twistable filaments such as DNA [31-33] (see Methods and SI).

This statement needs to be revised as it ignores a substantial body of work on self-organization of active filaments: (R. E. Isele-Holder, J. Elgeti, G. Gompper, *Soft Matter* 2015; Pfreundt et al, *Science* 2023; Faluweki et al *PRL* 2023; Kurjahn et al *Nature Comm* 2024).

Last point: the authors often say that their observations are reproducible ('...reproducibly forms macroscopic granules...'). What is meant? Different experiments on different days, different aliquots?

<https://doi.org/10.7554/eLife.112517.1.sa2>

Reviewer #3 (Public review):

Summary:

The authors report and characterize the formation of aggregate microstructures by the motile filamentous cyanobacterium *Fluctiforma draycotensis*, which exhibits gliding motility accompanied by rotation along the long axis while excreting EPS. In experiments with motile *F. draycotensis* cultures, they observed the formation of granular structures composed of cyanobacteria and other material (iron, polystyrene beads, etc.), with macrostructures on the scale of 1mm within 24 hours. The structures were motile at speeds comparable to that of the cyanobacteria filaments, resulting in their growth through coalescence over time. Notably,

such macrostructures were absent in nonmotile *F. draycotensis*, pointing to the role of filament motility in their formation. Through experiments examining the micro-scale dynamics, inert material such as small polystyrene beads was found to be transported by the gliding, buckling, and plectoneme dynamics of the filaments, pointing to the underlying mechanism by which particles are collected into larger-scale microgranule structures.

To interrogate the properties that drive the cyanobacteria filament buckling, plectoneme formation, and entanglement, the authors develop a mechanical model for filaments as nearly inextensible, slender bodies with resistance to twisting and bending under active gliding forces and torques and responding to fluid flows and surface adhesion. They derive expressions for the thresholds for buckling and twisting instabilities, which are additionally demonstrated and interrogated through simulation via the Immersed Boundary Method. Most importantly, bending and plectoneme formation only occur with sufficiently long filaments, and the threshold is shorter for bending than for plectoneme formation. Experimental observations with wild-type filaments agree with the model-predicted thresholds. The authors perform additional experiments with shorter filaments below both thresholds, including the filamentous bacterium *Pseudanabaena*, which fail to collect particles (though can in principle form macrostructures).

Strengths:

This work appears to be novel (notably, the discovery and characterization of the particle collection behavior of a filamentous cyanobacterium) and has interesting implications for both naturally observed cyanobacterial macrostructures as well as the controllable parameters in engineering them. The experimental and modeling work is well motivated, contributing to the broader understanding of macrostructure formation and material aggregation through active filament dynamics (not exclusive to cyanobacteria), as well as the underlying physical properties governing important filamentous cyanobacterium dynamics. As such, I would expect the results of this paper to be of broad interest to both biophysicists and microbiologists. Generally, the manuscript is well written with clear, compelling figures that illustrate the important conclusions of this study.

Weaknesses:

In the section on "Shorter gliding filaments cannot collect particles nor form granule macrostructures", the filamentous cyanobacteria considered *all* fall below the predicted thresholds for bending and twisting. The "long" *F. draycotensis* are 60 microns in length, notably less than the 120 and 320 micron thresholds derived in the previous section as well as the lengths of filaments considered in Figure 3D, yet these "long" 60 micron filaments form macrostructures. How can this be understood in the context of the model predictions? Is the nature of the macrostructures in Figure 4B, the microscale parameters, or the collection of particles somehow different than those with filaments an order of magnitude longer in earlier parts of the paper? The paper would be stronger if these sorts of questions were addressed in the text and/or with supplementary figures.

<https://doi.org/10.7554/eLife.112517.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary.

*In this manuscript, the authors investigate the mechanisms underlying macrostructure formation in a freshwater filamentous cyanobacterium strain, *F. draycotensis*, focusing*

on how its ability to aggregate and form these structures depends on the physical properties of the filaments. Using experimental observations, they demonstrate that the cyanobacterium actively captures and surrounds particles, a process driven primarily by gliding motility.

To explain these physical dynamics, the authors present a 3D model indicating that particle collection relies on filament length, as well as a specific mechanical response, namely, filament buckling and the subsequent formation of loops of bundles of filaments. While the authors have previously documented the buckling and looping characteristics of this strain, this study provides new insight by demonstrating that these physical phenomena are essential for particle capture and collection.

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This manuscript benefits from a rigorous and detailed quantitative analysis of video recordings, which clearly documents the motility, buckling behaviour, and particle collection dynamics of the filaments.

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To further confirm the length dependency within the same species, the authors experimentally generated shorter filaments of *F. draycotensis*. The fact that these shortened filaments also lose the capacity to collect particles provides strong evidence supporting their proposed mechanism.

We thank the reviewer for the accurate summary of our work and for their identified strengths of the study.

Weaknesses:

There is a conceptual concern. The authors linked the specific physical properties of this strain to evolutionary data, highlighting that the studied lineages diverged approximately two billion years ago. This creates a misleading impression that particle collection via flexible looping filaments is a recent evolutionary adaptation. However, particle collection has been observed in other cyanobacteria, such as *Trichodesmium*, which features short, rigid filaments. Therefore, the term “emerging” does not seem appropriate for the title and text. The capacity to collect particles in the studied strain *F. draycotensis* appears to be primarily a function of physical characteristics (filament length and flexibility) rather than evolutionary age. Any cyanobacterial strain possessing similar physical properties is likely to exhibit comparable behaviour, rendering the evolutionary timeframe largely irrelevant to the core mechanism.

We would like to first clarify our use of the term “emergent”. It seems that the referee took this in an evolutionary context, whereas we are using this term in the context of its use in systems dynamics, and referring to: “a complex entity displaying behaviors that its components do not have on their own, and emerge only when they interact in a wider whole”. Here, particle collection and dynamic aggregate formation “emerges” from the buckling and interaction of many filaments.

With regards to the evolution of particle collection behavior, our comment on the evolutionary distance between *F. draycotensis* and *Pseudoanabena* sp. was meant to highlight the point that particle collection seems to be a function of physical characteristics and motility: Despite a large evolutionary distance, and possibly many biological differences, a physics-based argument is capturing the difference between the particle collection ability of

these two organisms. Thus, we are in agreement here with the reviewer. We did not intend to make any arguments about “evolutionary age” of the particle collection behavior.

We see that the short, evolutionary comment in the Introduction has confused the reviewer and potentially is confusing to other readers too. We will therefore remove this evolutionary comment from the Introduction section of the revised manuscript and make the point in more detail in the Discussion section.

In addition, the phylogenetic tree presented in Figure S5 does not reflect the current consensus on cyanobacterial evolution and systematics and does not align with modern phylogenomic frameworks (see, for example, Strunecky et al., 2023 <https://doi.org/10.1111/jpy.13304>). There is also no such order Cyanobacteriales, which has been mentioned in a few older publications but is clearly outdated.

We thank the reviewer for this comment, as it has made us realise that we never explained our choice of taxonomic framework in the manuscript, and perhaps this is the source of the confusion.

The order Cyanobacteriales does exist: it is the order-level name applied in the Genome Taxonomy Database (GTDB) [5, 6], currently the most comprehensive and actively curated genome-based taxonomy of prokaryotes. GTDB classifies taxonomic groups algorithmically, as monophyletic groups in a concatenated marker-protein phylogeny with ranks normalised by relative evolutionary divergence. This has produced a number of re-groupings and new names relative to the older, morphology-derived classifications; many of these have since been formally proposed under the International Code of Nomenclature of Prokaryotes and the SeqCode [2], and are progressively being adopted by the NCBI. The placement of Cyanobacteriales, and of the other orders shown in Figure S5, can be inspected directly on the GTDB “Taxonomy Tree” (see here for the orders within the class Cyanobacteriia).

We would also like to note that we do not see our tree and the framework of Strunecky et al. as being in conflict. Strunecky et al. constructed their phylogenomic backbone using GTDB-Tk and the same 120-marker concatenated alignment that the GTDB itself uses. What differs between the two schemes is therefore not the underlying phylogeny but the nomenclature applied to the resulting clades: Strunecky et al. work within the botanical tradition and combine the phylogenomic tree with phenotypic characteristics, thereby proposing ten new orders and fifteen new families, whereas GTDB assigns rank boundaries purely by evolutionary divergence and so draws broader order limits. In practice, the GTDB order Cyanobacteriales spans several of the families (e.g. Oscillatoriales and Coleofasciculales) and orders (e.g. Chroococcales and Nostocales), that are proposed within the Strunecky et al. work. Our reason for adopting the GTDB nomenclature is for practical reasons specific to this study. *F. draycotensis* is a recently described organism [3] that is not included in Strunecky et al. and has no placement in their tree. In GTDB it falls within a family-level lineage (placeholder name JAAUUE01) inside the Cyanobacteriales, with the sequenced members of the Coleofasciculaceae as its closest relatives. We could not have assigned it to one of the Strunecky orders without inventing a placement. The same applies to some of the other, recent metagenomically described cyanobacteria [10], which similarly have no assigned names in the literature. GTDB, by contrast, provides a reproducible, algorithmic assignment for all of these genomes, and is now widely used for this reason in genome- and metagenome-based studies of cyanobacteria (e.g. [1]). We therefore used it consistently throughout.

Finally, with regards to the reviewer’s point about the tree itself, we would like to note that Figure S5 was intended only to convey the evolutionary distance between *F. draycotensis* and *Pseudanabaena* sp., and it was built from a modest set of six concatenated ribosomal protein markers using an approximate maximum-likelihood method with SH-like local support values. This is considerably less rigorous than the 120-marker RAxML and Bayesian analysis of Strunecky et al., and we agree that a stronger tree may be preferable. For the revised

manuscript we are recomputing the tree from a substantially larger set of concatenated single-copy marker genes, using IQTREE with model selection and non-parametric bootstrap support. We would note, however, that the specific conclusion drawn from this figure — that the two strains we use for our experimental work, namely *F. draycotensis* and *Pseudoanabena* sp. belong to deeply divergent cyanobacterial lineages — is supported by the deep backbone of the cyanobacterial tree, which is stable across marker sets and inference methods, and is equally supported by the tree of Strunecky et al.

We will make these points clearer in the Methods and Discussion sections of the revised manuscript, as well as the Figure S5 legend.

Another concern is that the authors nearly completely ignore the role of type IV pili in the gliding motility of cyanobacteria, including filamentous strains. For a long time, there was a misconception that the gliding motility of cyanobacteria was due to slime protrusion. Slime plays a role in this process. However, several studies have shown that filamentous strains also use type IV pili to glide on surfaces. The authors should discuss this and include it in their model.

The reviewer is correct that we did not include molecular details of gliding motility in our biophysical model. They are also correct to point out that pili and slime biosynthesis genes are shown to be involved in gliding motility [8]. It is, however, still unclear how these factors interact to produce mechanical gliding forces that can result in filament rotation (observed only in some filamentous cyanobacteria), filament reversal, as well as decoordination during such reversals, which we have previously shown in *F. draycotensis* [9]. Therefore, we have chosen to keep the biophysical model at a coarse-grained, phenomenological level. Instead of explicitly modelling the detailed molecular mechanisms behind force generation, we model only the minimum necessary physical forces and torques needed to reproduce the observed rotation and translation of the filament during gliding under de-coordinated conditions. This model is able to reproduce the experimentally observed buckling and twisting of filaments, and is therefore sufficient and useful to achieve a coarse-grained understanding of mechanical forces and their relationship to buckling, twisting and entanglement, which are the main processes we focus on here. As molecular details behind force generation in rotating, filamentous cyanobacteria become available, more detailed physical models can be constructed. We also note, in this context, that the two filamentous cyanobacteria we compare both encode the type IV pilus machinery, so the presence of a pilus motor does not by itself distinguish a particle-collecting from a non-collecting strain (see our response to the reviewer's next point).

We will make these points clearer in the Methods and Discussion sections of the revised manuscript.

*In addition, the authors concluded that gliding motility is responsible for particle collection by *Fluctiforma draycotensis*. Although I believe that their conclusion is correct, there might be several limitations to the experiments which allow for other reasons to be considered. Their conclusions were based on the use of a non-motile strain and an unspecified community without the motile *Fluctiforma draycotensis* strain. The problem I see here is that it is not clear why this strain is not motile; it could be because of the lack of type IV pili, mutations which alter their functionality, defects in slime secretion, any other mutation (e.g. in chemoreceptors), cellular structure, metabolism, or combinations of these. Furthermore, it is possible that the community changes its composition and behaviour when it lives without the cyanobacterium with a rich carbon source (glucose) or with a non-motile cyanobacterium which may not secrete slime or, for example, a signalling component which controls behaviour of the bacteria in the community. For that reason, the authors should be more cautious with their conclusion that solely motility behaviour of *Fluctiforma draycotensis* is responsible for particle collection. Additional factors might be responsible for these effects.*

Our conclusion that gliding motility is the main factor underpinning particle collection is based on several observations.

Firstly, on the macroscopic scale we present several control experiments where we did not observe particle collection: (i) in the community featuring a non-motile *F. draycotensis*, and with mostly the same other bacterial species as the community featuring the motile *F. draycotensis*, (ii) in a bacterial community derived from the original *F. draycotensis* community but lacking any cyanobacteria, (iii) in the original community with physically shortened *F. draycotensis*, and (iv) in another cyanobacterial community featuring different bacteria and a naturally shorter, filamentous gliding cyanobacteria *Pseudanabaena* sp. A straightforward, parsimonious explanation that satisfies all these observations is that particle collection is underpinned by physical characteristics of gliding filamentous cyanobacteria.

Secondly and more directly, in time-lapse microscopy imaging we repeatedly observe clusters of beads being moved by gliding filaments, and thereby being collected into larger clusters. Thus, whilst factors such as slime secretion also contribute, the primary mechanism driving the observed particle motion seems to be that particles stick to filaments and are carried around with them as they glide. We cannot rule out a contribution of pili to bead attachment and transport. We note, however, that both cyanobacteria compared here encode the type IV pilus machinery. In a homology survey of the two genomes, *Pseudanabaena* sp. and *F. draycotensis* both carry orthologues of the core T4P components — the assembly ATPase PilB, the retraction ATPase PilT, the inner-membrane platform protein PilC, the prepilin peptidase PilD, and the alignment-complex proteins PilM and PilF — together with the hormogonium-associated *hmpD*, *hmpF* and *hmpG*. *Pseudanabaena* sp. is therefore not pilus-deficient, and it does glide, yet it does not collect particles. The difference between the two organisms consequently cannot be attributed to the presence or absence of the pilus motor, which we would argue supports the physical argument we make here. Consistent with this, we have not identified mutations in pilus-related genes in the mutant, non-motile *F. draycotensis*.

We are currently in the process of preparing another manuscript describing the mutations that led to motility loss in the non-motile *F. draycotensis*, as well as the proteins that are differentially expressed in the motile and non-motile *F. draycotensis*. These analyses will shed more light on the molecular mechanisms abolishing motility and how they might be influencing particle collection.

In the revised manuscript, we will make these points clearer in the Discussion section.

Reviewer #2 (Public review):

Summary:

*The authors studied aggregation, buckling, and particle collection by the filamentous cyanobacterium *Fluctiforma draycotensis*, as well as by the filamentous *Pseudanabaena* sp. (order Pseudoanabinales). They performed a range of experiments, from imaging individual gliding filaments to multiple-day experiments showing the formation of large aggregates around a particle formed from a precipitate. They also developed a model of buckling filaments to argue that the ability of elastic filaments to collect particles and form macrostructures is confined to a part of the filament phase space in terms of length and flexibility, meaning that gliding combined with certain filament length and flexibility naturally reproduces the observations.*

Strengths:

This is an impressive study that uses multiple tools to connect macrostructure formation with filaments' gliding motility and buckling. It adds an important perspective on the biological and physical factors at play in the emergence of aggregates.

We thank the reviewer for the accurate summary of our work and highlighting the strengths of the study.

Weaknesses:

The authors ignore the possibility that filament behavior plays an important role in the emergence of the observed patterns. Cyanobacteria have been shown to control their gliding motility (Pfreundt et al Science 2023; Kurjahn et al Nature Comm 2024), and their molecular motors are known to be regulated by chemotaxislike signaling pathways (Risser ARM 2025). As far as I know, how the coordination between the pulling agents along an individual filament works is actively debated, but there seems to be little doubt that it exists.

*To illustrate this point better, note that the aggregation observed by the authors is consistent with the length-dependent ability of filaments to coordinate gliding (I'm not saying this is how it works in *Fluctiforma draycotensis*; I'm saying it's consistent). Suppose the coordination requires sufficiently long filaments, which could be the case when signaling molecules travel along the filament, propagating information about when individual pulling agents should reverse. In such a model, short filaments act randomly because they fail to coordinate gliding by the time they glide off nascent aggregates, whereas longer filaments can perform informed reversals because they have more time for coordination. Such behavior then explains the lack of aggregation in *Pseudanabaena* sp. (via behavior, not lack of stiffness). Note that *Trichodesmium* is stiff; its filaments do not buckle, yet *Trichodesmium* forms organized aggregates via tightly controlled motility. Note also that, as the authors report, since *Pseudanabaena* sp. is both shorter and faster, its filaments have relatively (to the time needed to glide the filaments' length) little time to coordinate reversals. In my opinion, whether the observed patterns passively emerge from gliding and buckling or result from active behavior remains an open question.*

We appreciate the comment by the reviewer. We certainly agree that behavioral responses exist in filamentous cyanobacteria and will interplay with the physical aspects to produce exciting, complex dynamics. Besides the exemplar ideas that the reviewer provides, there can be many other scenarios involving behavioral responses, such as responses to light and to quorum sensing molecules or photosynthesis-generated radicals. For example, in *F. draycotensis* we have observed photo-responses at the aggregate level, which we are currently studying. Photoresponses are also observed in *Trichodesmium* aggregates [7]. In

general, a full understanding of the interaction of the biological (i.e. behavioral) and the physical aspects will require several future studies.

In the current study, however, we focus on characterising the physical aspects of gliding motility alone, combined with experimental observations. We believe that this approach is important to establish a form of “null expectation” from the physics of gliding, elastic filaments alone. Currently, the molecular mechanisms responsible for coordinating the reversal behaviour of multiple filaments are still unclear, so it is difficult to experimentally demonstrate behavioural contributions to aggregate formation, e.g. via experiments where such behaviour is switched off. In the meantime, simulations such as those presented here allow us to test more precisely the potential role of activity, coordinated reversals and the elastic properties of the filament. In future it will be interesting to scale up the presented model to include multiple interacting filaments, and to systematically test the respective roles of active coordination behaviour for one individual filament (reversals) and for multiple interacting filaments (where contacts modulate activity), as well as the physical properties (length and flexibility). Such modelling studies can then identify if a ‘purely physical’ model can or cannot generate realistic aggregates, and pinpoint whether additional coordination mechanisms are needed to regulate aggregation. By testing the combination of different physical and biological coordination mechanisms, it would then help to indicate how much of a role is played by various potential active coordination behaviours.

We will bring out this point more clearly in the Discussion section of the revised manuscript.

I also have a small suggestion regarding this statement on model novelty:

The essential novelty of this model is that the filament itself is active and out of equilibrium, and additionally, the forces and torques are applied locally along its centreline, and not at its extremities as in previous steady-state mechanical studies of elastic, twistable filaments such as DNA [31-33] (see Methods and SI).

This statement needs to be revised as it ignores a substantial body of work on self-organization of active filaments: (R. E. Isele-Holder, J. Elgeti, G. Gompper, Soft Matter 2015; Pfreudnt et al, Science 2023; Faluwiki et al PRL 2023; Kurjahn et al Nature Comm 2024).

We agree with the reviewer that there is a significant literature on active filaments, some of which we have already cited and will now discuss in more details, as well as adding and discussing the suggested additional references. Our statement on “model novelty” refers to the analysis of buckling instabilities of biological filaments, and in particular DNA, due to a combination of forces and torques. To our knowledge, this has only been studied explicitly by [4], and only in the local (resistive force theory) limit. The elastohydrodynamic simulations coupled to local active forces and torques, as we implemented here, are therefore novel and will expand the analysis of both microbial filaments and other biological polymers. We will clarify these points in the Methods and Discussion sections of the revised manuscript.

Last point: the authors often say that their observations are reproducible (‘...reproducibly forms macroscopic granules...’). What is meant? Different experiments on different days, different aliquots?

The “replicability” statement was in reference to different experiments started on different days using cultures obtained from serial transfer experiments, as well as cultures re-initiated from cryostocks. This point will be made clear in the revised manuscript.

Reviewer #3 (Public review):

Summary:

*The authors report and characterize the formation of aggregate microstructures by the motile filamentous cyanobacterium *Fluctiforma draycotensis*, which exhibits gliding motility accompanied by rotation along the long axis while excreting EPS. In experiments with motile *F. draycotensis* cultures, they observed the formation of granular structures composed of cyanobacteria and other material (iron, polystyrene beads, etc.), with macrostructures on the scale of 1mm within 24 hours. The structures were motile at speeds comparable to that of the cyanobacteria filaments, resulting in their growth through coalescence over time. Notably, such macrostructures were absent in nonmotile *F. draycotensis*, pointing to the role of filament motility in their formation. Through experiments examining the micro-scale dynamics, inert material such as small polystyrene beads was found to be transported by the gliding, buckling, and plectoneme dynamics of the filaments, pointing to the underlying mechanism by which particles are collected into larger-scale microgranule structures.*

*To interrogate the properties that drive the cyanobacteria filament buckling, plectoneme formation, and entanglement, the authors develop a mechanical model for filaments as nearly inextensible, slender bodies with resistance to twisting and bending under active gliding forces and torques and responding to fluid flows and surface adhesion. They derive expressions for the thresholds for buckling and twisting instabilities, which are additionally demonstrated and interrogated through simulation via the Immersed Boundary Method. Most importantly, bending and plectoneme formation only occur with sufficiently long filaments, and the threshold is shorter for bending than for plectoneme formation. Experimental observations with wild-type filaments agree with the model-predicted thresholds. The authors perform additional experiments with shorter filaments below both thresholds, including the filamentous bacterium *Pseudanabaena*, which fail to collect particles (though can in principle form macrostructures).*

Strengths:

This work appears to be novel (notably, the discovery and characterization of the particle collection behavior of a filamentous cyanobacterium) and has interesting implications for both naturally observed cyanobacterial macrostructures as well as the controllable parameters in engineering them. The experimental and modeling work is well motivated, contributing to the broader understanding of macrostructure formation and material aggregation through active filament dynamics (not exclusive to cyanobacteria), as well as the underlying physical properties governing important filamentous cyanobacterium dynamics. As such, I would expect the results of this paper to be of broad interest to both biophysicists and microbiologists. Generally, the manuscript is well written with clear, compelling figures that illustrate the important conclusions of this study.

We thank the reviewer for the accurate summary of our work and recognising the broad relevance of the study.

Weaknesses:

*In the section on “Shorter gliding filaments cannot collect particles nor form granule macrostructures”, the filamentous cyanobacteria considered “all” fall below the predicted thresholds for bending and twisting. The “long” *F. draycotensis* are 60 microns in length, notably less than the 120 and 320 micron thresholds derived in the previous section as well as the lengths of filaments considered in Figure 3D, yet these “long” 60 micron filaments form macrostructures. How can this be understood in the context of the model predictions? Is the nature of the macrostructures in Figure 4B, the microscale parameters, or the collection of particles somehow different than those with filaments an order of magnitude longer in earlier parts of the paper? The paper would be stronger if these sorts of questions were addressed in the text and/or with supplementary figures.*

We thank the reviewer for this point. Indeed as we mention in the text, the ‘long’ population has a mean length of 60 micron. However, as shown in the length distribution plot in Fig 4A, the maximum filament lengths observed in these populations (within the samples used for microscopy) are 560 microns for the long filaments, versus 240 microns for the short filaments. Thus, we expect the long population to contain multiple filaments that can buckle and a few that can form plectonemes, whilst the short population might have some buckling filaments and none that form plectonemes. We stress that Fig 4A only shows the length distribution for what we believe to be a representative sample taken from the long and short populations, not the full data from the entire population.

We will revise the main text to include the maximum filament lengths of the two populations as well as the mean values. We will also add lines to Fig 4A to indicate the buckling and plectoneme threshold lengths from the analytical estimate for the *F. draycotensis* filaments (same values as in Fig 3), to make it clear that the long population contains more buckling/plectoneming filaments than the short population.

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